Internet Protocols Explained

Internet protocols refer to the rules required by different applications for the exchange of data over the internet. In layman terms, they are like languages. Just as language is used as a medium of communication, a standardised protocol is necessary to communicate across different computers using different hardware and software. This first article in a two-part series focuses on the internet protocols used till the network layer.

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The internet protocol (IP) address is a unique number allocated to a domain or device by which other devices identify it in a network. There are two versions of this IP address:

- Addresses in IPv4 are 32-bits, 2 power 32; example: 12.244.233.165
- Addresses in IPv6 are 128-bits, 2 power 128; example: 2001:0db8:00 00:0000:0000:ff00:0042:7879

You may ask that if the IP address serves as a unique identity and so does the MAC address, can we use the MAC address itself? Well, we can, but it would take a very long time to search and find the destination. In the case of an IP address, the International Assigned Numbers Authority (IANA) assigns both the IPv4 and IPv6 addresses in a hierarchical manner. IANA assigns blocks of IP addresses to regional internet registries. The regional registries in turn assign smaller blocks to national registries, and so on, with blocks eventually being assigned to individual internet service providers (ISP). It's the ISPs that assign specific IP addresses to individual devices, and there are a couple of ways they can do this. As you can see, it's easy to find the IP address from the tree and get its geo location.

Ports

Ports are used for each application that is used for communication. They allow computers to differentiate between the different kinds of traffic such as web, email, file, voice, etc. Ports are represented by 16-bit numbers. There are 2^16, i.e., 65336 port numbers and they come in three ranges.

Well-known port numbers: 0 to 1023 are well-known port numbers as they are used by well-known protocol services. These are allocated to server services by the IANA, which is a division of ICANN (Internet Corporation for Assigned Names and Numbers). ICANN is a non-profit organisation that was established in the United States in 1998 to help maintain the security of the internet and allow it to be usable by all.

Registered port numbers: 1024 to 49151 are registered port numbers, i.e., these can be registered to specific protocols by software corporations.

Dynamic port numbers: 49152 to 65536 are dynamic port numbers and they can be used by anyone.

Let's try and understand these terms in a simple way. If I want to send a message to Han who understands only Japanese, Japanese is my protocol to communicate with him, his home address is the IP address, and the letter drop box number where he collects all the post is the port number.

OSI model

The OSI (open system interconnection) model provides a standard for different computer systems to communicate with each other. Figure 1 explains each layer.

Let's take a simple example and see what happens in each layer when you hit *google.com.*

- In the **application layer**, the HTTP protocol is used and messages are formed. This is the only layer that interacts with the user.